

CAPSTONE RESEARCH GAP REPORT

Poison-Resilient, Novelty-Aware Continual Learning for Transformer-Based Network Intrusion Detection

Strengthened working direction for a 6-credit engineering capstone

Executive verdict

The secure replay / poisoning-defense direction is stronger than the earlier attention-shift/drift idea for this capstone. However, the strongest version is not merely “add a poison filter.” The defensible research problem is to secure replay-based continual NIDS while preserving the ability to learn genuinely novel attacks - a security-versus-plasticity problem that must be evaluated against label-flip and stealthy backdoor poisoning, class imbalance, continual-learning forgetting, and multi-dataset generalization.

Prepared for Review 1 - 21 August 2026

Rapid, targeted literature synthesis using recent peer-reviewed work and current 2026 preprints.

Novelty status: strong candidate gap, not a “first-ever” claim. A publication submission should still run a reproducible IEEE/ACM/Scopus/Web of Science novelty search.

1. Decision: Which Idea Is Better?

Two candidate directions were compared: (A) attention-shift-guided continual intrusion detection, where attention distributions help signal concept/feature drift; and (B) secure continual learning, where a replay-based Transformer NIDS is defended against poisoning of the replay/update pipeline.

Criterion	A: Attention/drift idea	B: Secure replay idea
Research crowding in 2026	Drift-aware NIDS, attention + incremental learning, DRIFT-CL, SSF, RepShield and related methods already occupy much of the space.	Replay poisoning in a continual Transformer NIDS was only explicitly stress-tested in a very recent Aug-2026 preprint; defenses are explicitly left unevaluated.
Gap directness	Requires a subtle claim about using attention statistics as a drift signal; attention is also not automatically an explanation.	Clear failure chain: replay fixes forgetting -> replay becomes persistent trusted state -> poisoning persists across future updates.
Security contribution	Primarily adaptation/generalization research.	Adaptive ML + adversarial ML + secure model updating: clearly cybersecurity-centered.
Measurable experiments	Good, but drift definition/evaluation can become diffuse.	Very concrete attacks, defense rates, forgetting, new-class learning, ASR, latency, memory.
6-credit scope	Solid but novelty risk is higher.	Excellent: baseline reproduction + attacks + defense + ablation + secondary dataset + optional dashboard.
Publication headroom	Moderate.	Higher if the defense addresses the novelty-vs-poison ambiguity and is evaluated rigorously.
Main risk	A nearby 2026 paper may already cover the exact drift trigger.	Generic CL/NIDS poisoning defenses exist, so exact defense cannot be a naive outlier filter or generic sanitizer.

Recommendation

Choose Idea B, but strengthen it to: “Poison-Resilient, Novelty-Aware Secure Replay for Continual Transformer-Based Network Intrusion Detection.” Keep attention/Transformer subject knowledge as the detection backbone, while the thesis contribution targets secure continual adaptation.

Why the earlier idea loses: by 2024-2026, attention-based incremental IDS, drift detection with continual learning, strategic replay/forgetting, and attention-based feature analysis are all active areas. DRIFT-CL, for example, already combines drift monitoring, importance-weighted replay and attention-based feature importance. Therefore, attention + drift can still be studied, but it has less novelty headroom for a rushed 6-credit capstone.

2. The Evidence Chain Behind the Better Idea

The strongest recent anchor is Ariffin et al. (arXiv:2608.04602, 5 Aug 2026). It couples a tabular Transformer with benign-anchored class-balanced replay and evaluates both conventional class-incremental (CI) and a more realistic class-instance incremental (CII) scenario.

- Under clean replay, the reported continual learner approaches joint-training performance on CICIDS2017, with very low forgetting.
- Under replay-buffer label flipping, the model can collapse catastrophically; the paper reports accuracy falling to roughly 0.0053 at its 1% buffer-budget setting.
- Under a stealthy inter-arrival-time backdoor, aggregate accuracy can remain above 0.97 while triggered attack success reaches about 95-100%, so a normal accuracy dashboard may not expose the corruption.

- The paper explicitly lists defenses against the label-flip/backdoor attacks as unevaluated future work, and also calls for CII attack evaluation, additional datasets, adaptive triggers, buffer auditing, and longer experience streams.

Why this matters for your capstone

A recent paper has done the vulnerability demonstration and has effectively handed you a set of open follow-up questions. That is much safer than inventing a vague “research gap.” Your capstone can replicate the vulnerable baseline, then investigate a defensible NIDS-specific protection mechanism.

3. But What Is Already Known? - Avoiding a False Novelty Claim

The gap is not “data poisoning has never been studied,” “continual learning is new,” or “replay protection is new.” All three claims would be false.

Established area	Examples located	Implication for this project
Continual NIDS + replay	NeurIPS 2023 augmented-memory replay; SPIDER 2024; SSF INFOCOM 2025; TAMR/Adaptive Memory Replay 2025; ACIL and RepShield 2026.	Replay and continual NIDS are established. Your novelty cannot be “use replay.”
Poisoning continual learners	PACOL; persistent continual-learning backdoors; targeted forgetting/false-memory attacks; 2026 theoretical poisoning analysis.	Vulnerability is established outside NIDS as well.
Poisoning defenses	PACOL tests sanitization/outlier defenses; Pawlak et al. propose poisoned-task detection/defense; theoretical task verification exists; generic robust/noise-tolerant replay research exists.	A confidence threshold or Isolation Forest alone is unlikely to be novel.
Poisoning defenses for NIDS	SecFedNIDS protects federated NIDS; Monkam et al. use topological data analysis for early poison detection in ML-NIDS.	NIDS-specific poisoning defense exists in other learning settings; distinguish continual replay.
Avoid learning malicious behavior	METANOIA explicitly identifies a “discrimination paradox” in lifelong provenance-based intrusion detection.	The high-level malicious-vs-benign memory problem is not unique; your setting is flow-based supervised class-incremental replay under adversarial poisoning.

4. Refined Research Gap

Candidate gap statement - use this wording

Replay-based continual learning is increasingly effective for adaptive network intrusion detection, but replay memory becomes persistent trusted training state. Recent work has demonstrated catastrophic label-flip and stealthy timing-backdoor poisoning against a replay-based Transformer NIDS and explicitly leaves defenses unevaluated. Generic continual-learning poison defenses and static/federated NIDS sanitization methods exist, but the literature located in this rapid review does not establish a lightweight, flow-NIDS-specific replay admission and re-auditing mechanism that is jointly evaluated for (i) class-incremental retention, (ii) learning genuine emerging attack classes, (iii) overt and stealthy replay poisoning, (iv) CI/CII traffic exposure, and (v) cross-dataset robustness. We therefore investigate this intersection rather than claiming poisoning or continual learning themselves are new.

The scientific difficulty is a stability-plasticity-security trade-off: the NIDS must remain plastic enough to learn a genuinely new attack, stable enough not to forget old attacks, and secure enough not to memorize adversarially corrupted update samples.

5. The Strongest Research Problem: Novel Attack or Poison?

A naive poison filter creates a contradiction. Continual learning is needed because new attacks differ from known traffic. But a defense that rejects every unusual sample will also reject the very attacks the system is supposed to learn.

Novelty-versus-poison ambiguity

Unfamiliar update sample -> (A) legitimate emerging attack that should be learned, (B) benign distribution change that should not become an “attack,” or (C) poisoned/backdoored update that should be quarantined. The research problem is not simply anomaly detection; it is safe admission of training evidence under limited trust.

An analogous “discrimination paradox” appears in METANOIA for lifelong provenance-based intrusion detection: if the detector already knows whether replay material is malicious, replay is unnecessary; if it does not, learning malicious behavior can harm detection. METANOIA is an important adjacent work, but its data representation, detection paradigm and threat model are different from a replay-poisoned flow-based Transformer classifier. The capstone should cite it rather than pretend the ambiguity has never been recognized.

6. Strengthened Project Definition

Working title: Poison-Resilient, Novelty-Aware Secure Replay for Continual Transformer-Based Network Intrusion Detection

Module	Role
Detection engine	Lightweight tabular Transformer / feature-token Transformer over flow-level fields. Attention remains directly aligned with the guide’s requested subject knowledge.
Continual learner	Class-balanced experience replay, preferably reproducing CI and CII protocols from the Aug-2026 anchor.
Secure replay admission	A lightweight multi-signal gate scores candidate samples before they become persistent training memory.
Quarantine / novelty pool	Uncertain samples are not immediately discarded. They accumulate temporarily so coherent new patterns can be distinguished from isolated poison and can receive limited analyst/oracle verification.

Module	Role
Periodic buffer audit	Stored exemplars are rechecked because a one-time admission decision may become invalid as representation and class statistics evolve.
Security attacks	Core: label flipping + timing-feature backdoor. Stretch: replay-eviction/sampler attack or clean-label backdoor.
Evaluation	Detection + continual learning + poisoning resistance + novelty learning + efficiency, under repeated seeds and at least one secondary dataset/cleaned variant.

7. Proposed Secure Replay Controller

Do not hard-code the final defense before a focused defense survey. A strong and feasible candidate is a multi-signal admission score with an explicit quarantine path. Each signal must be ablated so the thesis can show what actually contributes.

Signal	Purpose / limitation
1. Prediction/label consistency	If an externally supplied label strongly conflicts with a calibrated model prediction, mark the sample risky. Useful against simple label flipping, but weak for new classes and sophisticated backdoors.
2. Representation support	Measure distance from trusted class prototypes / robust class neighborhoods in the Transformer embedding. A label that is incompatible with its feature representation is suspicious.
3. Robust feature anomaly	Use a lightweight statistical/outlier baseline on selected flow features or latent representation. This is a comparator, not by itself the novelty.
4. Temporal/cohort persistence	A genuine emerging attack is more likely to form a repeated coherent pattern across a short window; isolated or inconsistent points are more suspicious. This helps avoid rejecting all novelty.
5. Optional gradient-conflict signal	Stretch: compare candidate gradient direction with trusted class/update gradients. Large persistent conflict can indicate harmful training influence but increases cost.
6. Buffer re-audit	Periodically rescore replay items and quarantine items whose label/representation consistency deteriorates.

Conceptual trust score (one implementation option):

$$\text{Trust}(\mathbf{x}) = w_1 \cdot \text{LabelConsistency} + w_2 \cdot \text{PrototypeSupport} + w_3 \cdot \text{CohortPersistence} - w_4 \cdot \text{AnomalyRisk}$$

Weights should not be arbitrarily declared “optimal.” They can be selected on a clean validation stream or learned with a very small calibration model, then held fixed during adversarial evaluation. A rule-based version is a valid fallback if learning a gate becomes too complex.

Important security distinction

Cryptographic hashes/HMACs can detect post-write tampering of a buffer file, but they cannot tell whether a semantically poisoned sample was legitimately admitted through the data/label pipeline. Storage integrity can be an engineering hardening feature; it is not the main ML poisoning defense.

8. Threat Model - Make the Security Claim Precise

A research-grade defense needs a defined attacker. Otherwise “secure replay” is too broad.

Element	Definition
Attacker goal	Untargeted collapse/forgetting, or targeted misclassification when a trigger appears.
Attacker capability - core	Can cause a bounded fraction of candidate update/replay samples to be mislabeled or feature-modified before/at admission. Does not directly edit model weights or gradients.
Attack 1: label flip	Change stored/candidate supervision so repeated replay pushes wrong decision boundaries.
Attack 2: timing backdoor	Perturb a controllable flow timing feature and associate it with a target label, seeking high attack success while clean metrics remain strong.
Defender knowledge	A small trusted seed/buffer for known classes, current model outputs/embeddings, and a limited analyst/oracle budget for ambiguous novelty.
Out of core scope	Full white-box gradient poisoning, model-weight compromise, network-stack exploitation, and arbitrary operating-system compromise.

This is safer than copying the anchor paper’s strongest “write access to stored exemplars” threat model without modification. For a practical system, semantic poisoning through an update/label pipeline is more interesting because a filesystem HMAC would trivially detect unauthorized byte-level modification but not trusted-channel bad data.

9. Research Questions and Testable Hypotheses

ID	Research question	Role
RQ1	How vulnerable is a clean replay-based Transformer NIDS under CI and CII to low-budget label-flip and stealthy timing-backdoor poisoning?	Reproduction/extension; not the novelty.
RQ2	Can a multi-signal replay admission controller reduce poisoned-sample admission and backdoor ASR without materially degrading clean macro-F1 and old-class retention?	Primary defense question.
RQ3	Can quarantine/cohort logic preserve learning of genuine newly introduced attack classes better than a naive outlier/confidence filter?	Primary novelty-vs-poison question.
RQ4	Does periodic buffer re-auditing add protection after admission, and what memory/update overhead does it impose?	Persistence question.
RQ5	Does the defense remain useful on a corrected/secondary dataset rather than only CICIDS2017?	Generalization/credibility question.

Example hypotheses: (H1) the proposed gate significantly reduces poison admission and ASR versus unprotected ER-Balanced; (H2) it causes substantially fewer false rejections of legitimate new-class samples than a generic outlier-only filter; (H3) its security gains remain after controlling for replay-buffer size and clean-data performance.

10. Experimental Protocol

1. Reproduce a clean baseline: tabular Transformer + class-balanced experience replay, with CI and CII task streams. Validate that forgetting and macro-F1 are in the expected range before adding defenses.
2. Implement poisoning harnesses: label flip and timing-feature backdoor at multiple bounded budgets. Report both aggregate degradation and targeted attack success.
3. Implement simple defenses first: calibrated confidence/label-consistency, Isolation Forest or robust distance, and class-prototype consistency.
4. Implement the proposed combined gate plus quarantine/novelty pool and periodic re-audit.
5. Run ablations: remove one signal at a time; compare admission threshold sensitivity; compare with/without quarantine and re-audit.
6. Evaluate on clean stream, poisoned stream and genuine-new-class stream. A defense that “wins” only because it rejects all new data fails the research objective.
7. Repeat with multiple random seeds and report mean $\pm$ standard deviation or confidence intervals where feasible.
8. Repeat the strongest baseline/defense on a corrected or secondary NIDS dataset to test whether the result is tied to CICIDS2017 artifacts.

11. Baselines You Need

Baseline	Method	Why include it
No continual learning	Static model / joint oracle (where appropriate)	Context / upper bound.
Naive sequential fine-tuning	No replay	Forgetting lower bound.
Regularization CL	EWC or LwF	Shows whether defense gains depend on replay.
Replay baseline	ER-Stratified / ER-Balanced	Main vulnerable baseline.
Naive security filter	Confidence threshold	Cheap baseline.
Generic outlier defense	Isolation Forest / LOF / robust distance	Tests whether a generic sanitizer is already sufficient.
Prototype consistency	Class-conditional embedding distance	Strong lightweight comparator.
Proposed	Multi-signal gate + quarantine + re-audit	Target method.

Optional advanced comparator: adapt an existing generic continual-learning poisoning/noise-tolerant replay defense if a clean implementation is available. This is desirable for publication, but not required to make the 6-credit capstone complete if implementation time is limited.

12. Datasets - Strengthened Beyond the Uploaded Draft

The uploaded concept used CICIDS2017 as the primary candidate. Keep it because the Aug-2026 anchor paper uses it, making reproduction possible, but do not let the thesis depend on CICIDS2017 alone.

Dataset	Use	Caution
CICIDS2017	Primary replication dataset. Public flow CSVs and PCAPs; direct comparability to the 2026 replay-poisoning anchor.	Known feature-extraction and labeling issues; very high benchmark scores should not be interpreted as production accuracy.
LYCOS-IDS2017	Corrected re-extraction of CICIDS2017 raw traffic using LycoSTand; useful robustness check.	Feature schema differs from the original; mapping may require care.
NF-CSE-CIC-IDS2018-v2 / NF-UNSW-NB15-v2	UQ NetFlow V2 datasets share a 43-feature schema; attractive for secondary-domain testing.	Adds preprocessing work; restrict to one secondary dataset if schedule is tight.
AnoShift	Long-horizon naturally shifting benchmark used in CL literature.	Useful stretch goal, but its anomaly-detection setting may not align cleanly with the supervised class-incremental poisoning protocol.

Recommended dataset plan

Phase 1: reproduce and attack on CICIDS2017. Phase 2: re-run the strongest defense on LYCOS-IDS2017 or one UQ NetFlow V2 dataset. Do not promise four datasets for Review 1; two well-designed evaluations are better than shallow experiments on many.

Why this matters: cross-dataset studies report near-perfect within-dataset NIDS performance but severe degradation when train and test distributions differ. Dataset quality studies also document errors in CICIDS2017 feature extraction and labeling. A secondary/corrected dataset therefore materially strengthens the thesis.

13. Metrics - Accuracy Alone Is Not Acceptable

Area	Metrics	Reason
Detection	Macro-F1, per-class precision/recall/F1, false-positive rate, PR-AUC where suitable	Handles severe class imbalance and rare attacks.
Continual learning	Average forgetting, old-class retention, intransigence/plasticity, backward transfer if implemented	Separates “never forgets” from “refuses to learn.”
Security	Attack success rate (ASR), poison admission rate, poison detection TPR/FPR, clean-sample rejection rate	Measures actual defense rather than clean accuracy.
Novelty	New-class acceptance/learning rate, false quarantine of legitimate novel samples, time/windows to accept a coherent novel pattern, analyst-query budget	Directly measures the novelty-vs-poison trade-off.
Efficiency	Replay MB, gate latency/sample, update time, inference latency, model size	Important for the “lightweight” claim.

14. Ablation Studies - Where the Thesis Becomes Research

- Confidence/label consistency only vs prototype distance only vs anomaly score only vs cohort persistence only vs combined gate.
- No quarantine vs quarantine; no re-audit vs periodic re-audit.
- Different buffer budgets and poison budgets.
- CI vs CII exposure.
- Transformer vs a matched MLP with the same replay/defense protocol, if time permits. The Aug-2026 anchor itself lists this architectural ablation as incomplete.
- Clean labels vs limited/uncertain labels if a semi-supervised stretch experiment is feasible.
- Defense threshold sensitivity: show the security/utility Pareto trade-off rather than choosing a magic threshold.

15. What Makes This a Solid 6-Credit Engineering Capstone?

Dimension	Why it qualifies
Research depth	Three interacting problems: class-incremental adaptation, catastrophic forgetting, adversarial poisoning.
Engineering depth	Dataset pipeline, Transformer baseline, replay manager, poisoning harness, trust gate, quarantine store, buffer auditor, experiment runner.
Scientific evaluation	Baselines, attacks, ablations, multiple seeds, secondary dataset, multi-axis metrics.
Cybersecurity relevance	The IDS update mechanism itself becomes a security boundary; model monitoring cannot rely only on clean accuracy.
Feasibility	Flow-level tabular data; no LLM-scale compute; public datasets; anchor implementation is publicly released by its authors.
Demo potential	Dashboard can show detections, buffer admissions, quarantined cohorts, update events, ASR/security metrics. Optional and secondary to research.

Scope rule

Do not waste the research semester on Raspberry Pi deployment, a polished React dashboard, or packet capture until the attack/defense experiments work. Those are presentation layers, not the thesis contribution.

16. Deliverables - Minimum, Strong, and Stretch

Scope	Deliverables
Minimum defensible	Clean Transformer NIDS; replay-based CL; CI task stream; label-flip poisoning; one replay defense; detection + forgetting + poison metrics; reproducible report.
Strong 6-credit target	CI + CII; label flip + timing backdoor; multi-signal admission; quarantine novelty pool; re-audit; ablations; two dataset settings; repeated seeds; efficiency analysis.
Stretch / paper extension	Long 10+ experience streams; replay-eviction/sampling attack; adaptive trigger; semi-supervised label budget; matched MLP ablation; automated analyst feedback; edge deployment.

17. Timeline

Period	Work
Weeks 1-2	Deep literature review: Transformer attention, continual learning, replay, NIDS poisoning; reproduce dataset preprocessing.
Weeks 3-4	Clean static + sequential + ER baselines; implement CI and CII stream construction.
Weeks 5-6	Reproduce label-flip and timing-backdoor attacks; validate ASR and forgetting.
Weeks 7-8	Implement confidence/outlier/prototype defense baselines.
Weeks 9-10	Implement combined secure replay gate + quarantine/novelty pool.
Weeks 11-12	Periodic buffer re-audit; ablations and threshold/security-utility curves.
Weeks 13-14	Secondary/corrected dataset; repeated seeds; statistical analysis; efficiency.
Weeks 15+	Paper/thesis writing, reproducibility package, final demo/dashboard if core work is stable.

18. Risk Register and Fallbacks

Risk	Severity	Mitigation
Defense is not novel enough	High	Do not lock mechanism now. Compare against PACOL sanitization, TDA-NIDS poisoning defense, SecFedNIDS, METANOIA, and generic robust replay; formulate contribution around replay-admission + novelty preservation + NIDS threat model/evaluation.
Backdoor too hard to detect	High	This is scientifically valuable. Report defense limits honestly; prioritize reduction in ASR with bounded clean utility. Add gradient or re-audit signal as stretch.

Risk	Severity	Mitigation
Outlier filter rejects new attacks	High	Use quarantine/cohort persistence and limited oracle verification; make this trade-off an explicit metric.
CICIDS2017 results look unrealistically perfect	Medium	Use macro-F1, attacks, corrected/secondary dataset, and discuss known dataset defects.
Compute/time limits	Medium	Use stratified subsets for development, one secondary dataset, small Transformer, fixed seed pipeline; reserve full runs for final experiments.
Too much application/UI work	Medium	Dashboard only after experiments; no hardware dependency for the core contribution.

19. Review-1 Material - Match the Faculty Rubric

19.1 Motivation and Objectives

Motivation: Static NIDS models become stale as new attacks emerge. Continual learning can update the detector without full retraining, but replay - one of the strongest anti-forgetting mechanisms - creates persistent training state. A newly released Transformer-NIDS study shows that replay corruption can either collapse performance or implant a stealthy backdoor while aggregate accuracy remains high. Therefore adaptive detection must be secured, not merely made accurate.

Objectives:

- Understand attention/Transformer feature representation for flow-level intrusion detection.
- Reproduce a lightweight replay-based continual NIDS under CI/CII task streams.
- Quantify replay poisoning using overt label flipping and a stealthy timing-feature backdoor.
- Design and evaluate a lightweight replay admission/auditing mechanism that reduces poison persistence while retaining genuine novel-attack learning.
- Evaluate detection, forgetting, security, novelty and efficiency, and validate the strongest result beyond the original CICIDS2017 setting.

19.2 Problem Survey

Problem survey conclusion: Transformer NIDS is not novel; continual replay NIDS is not novel; data poisoning is not novel; generic poison defenses exist. The open candidate intersection is secure replay/update admission for a flow-based continual Transformer NIDS under explicit poisoning and novelty-learning requirements. The most recent anchor explicitly leaves defenses and CII attacks unevaluated, while adjacent NIDS and generic CL works show why a trivial filter is insufficient.

19.3 Subject Knowledge

- Scaled dot-product self-attention: $\text{Attention}(Q,K,V)=\text{softmax}(QK^T/\sqrt{d_k})V$; multi-head attention learns multiple representation subspaces.
- Tabular Transformers tokenize/represent heterogeneous flow features and use attention to model feature interactions.
- Continual learning: stability-plasticity dilemma, catastrophic forgetting, replay, regularization (EWC/LwF), class-incremental vs class-instance incremental learning.
- Adversarial ML: training-time poisoning, label flipping, backdoors, attack success rate, clean-accuracy stealthiness.
- Security-vs-plasticity: defensive filtering must not convert an adaptive IDS into a static one by rejecting all unfamiliar data.
- Dataset integrity and class imbalance: why macro-F1, per-class recall and corrected/cross-domain validation matter.

19.4 Methodology and Timeline

Methodology: baseline reproduction -> continual replay -> controlled poisoning -> defense baselines -> proposed secure replay gate -> quarantine/re-audit -> ablation -> corrected/secondary dataset -> statistical/efficiency evaluation. The detailed timeline is in Section 17.

20. Guide Questions - Defensible Answers

Guide question	Answer
"Transformer IDS already exists. What is new?"	Correct. Transformer is our detection backbone and subject-knowledge component, not the novelty. The candidate contribution is protecting continual replay/update memory against poisoning while preserving new-attack learning.
"Continual-learning NIDS already exists."	Yes. NeurIPS 2023, SPIDER, SSF, ACIL, TAMR and RepShield establish that. We build on replay rather than claim it.

Guide question	Answer
“Poisoning defenses already exist.”	Yes. Generic CL defenses and static/federated NIDS poison defenses exist. Our gap is narrower: replay-specific continual flow NIDS, explicit backdoor/label-flip threats, CI/CII, and the novelty-admission trade-off.
“Why not just Isolation Forest?”	It is a necessary baseline, but a new attack is also an outlier. If the defense rejects novelty, it defeats continual learning. We need to measure this directly and use quarantine/cohort or class-conditional evidence.
“Is the Aug-2026 paper peer reviewed?”	No, it is a very recent arXiv preprint. We use it as the immediate vulnerability anchor, but ground continual NIDS and poisoning context in peer-reviewed NeurIPS/INFOCOM/Computer Networks/PMLR/USENIX and other published work.
“Why CICIDS2017?”	For direct reproduction of the anchor. But it has known extraction/label issues, so we plan a corrected/secondary dataset check rather than treating its accuracy as production evidence.
“What exactly will you prove?”	Empirically test whether the proposed secure admission/audit method improves the security-utility-plasticity trade-off: lower poison admission/ASR, preserved macro-F1 and old-class retention, and preserved learning of legitimate new classes at reasonable cost.

21. Priority Reading List

Priority	Paper	Why
READ FULL	Ariffin et al. 2026 - Adaptive IDS using Transformer + continual learning + adversarial replay investigation	Your direct anchor; focus Sections 2.3-2.5, 3.4, 3.8-3.9, 5.5-5.6, 7.
READ FULL/CORE	Amalapuram et al. 2023 - Augmented Memory Replay NIDS (NeurIPS)	Establishes replay-based continual NIDS and class imbalance/scalability.
READ CORE	SSF, INFOCOM 2025	Shows strategic sample selection/forgetting and refreshed memory under drift already exist.
READ CORE	PACOL, 2023	Shows CL poisoning and tests sanitization/outlier defenses; prevents a naive novelty claim.
READ CORE	Pawlak et al., CoLLAs/PMLR 2026	Peer-reviewed recent poisoning/defense work in exemplar-free CL.
READ CORE	Persistent Backdoor Attacks in Continual Learning, USENIX Security 2025	Shows persistent backdoors across continual updates and defense evasion.
READ CORE	Monkam et al., Computers & Security 2024	NIDS-specific data-poison detection using TDA; important defense comparator.
SKIM	METANOIA 2026 version	Adjacent lifelong PIDS; explicitly addresses not learning malicious behaviors / “discrimination paradox.”
SKIM	ACIL 2026; RepShield 2026; TAMR 2025; SPIDER 2024	Maps current adaptive NIDS landscape.
SKIM	LYCOS-IDS2017 + cross-dataset generalization	Dataset-quality and generalization rationale.

22. Rapid Literature Search Method and Novelty Confidence

This report used a rapid, targeted literature search rather than a formal PRISMA systematic review. Searches were focused on current publisher/arXiv/PMLR/USENIX/IEEE-discoverable material and included combinations of: “continual learning network intrusion detection replay,” “replay buffer poisoning continual learning,” “data poisoning continual network intrusion detection defense,” “Transformer continual NIDS,” “concept drift attention continual IDS,” “replay buffer integrity NIDS,” and “CICIDS2017 corrected dataset.”

Inclusion emphasis: works directly addressing (i) continual NIDS, (ii) experience replay/memory, (iii) poisoning/backdoors in continual learners, (iv) poisoning defense in NIDS, (v) drift/attention approaches that compete with the earlier idea, and (vi) dataset integrity/generalization. Recent 2026 papers were retained even when preprints, but are explicitly labeled as such.

Confidence assessment

Project-worthiness: HIGH. 6-credit feasibility: HIGH. Research-gap confidence: MODERATE-HIGH as a candidate intersection. Publication novelty: NOT YET GUARANTEED. Before writing “first” or “novel” in a paper, run a documented IEEE Xplore + ACM DL + Scopus/Web of Science/Google Scholar search and compare the exact defense mechanism, threat model and evaluation protocol.

23. References and Source Map

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24. One-Page Proposal Summary

Working title

Poison-Resilient, Novelty-Aware Secure Replay for Continual Transformer-Based Network Intrusion Detection

Problem: Replay-based continual NIDS can retain old attacks while learning new ones, but replay memory becomes persistent training state. Recent work shows replay poisoning can either collapse the detector or implant a stealthy timing backdoor while clean accuracy remains high.

Gap: Defenses are not absent in the broader literature, but the exact combination of replay-specific protection, flow-based Transformer NIDS, genuine novel-class learning, CI/CII evaluation, overt + stealthy poisoning, and secondary-dataset validation remains a strong underexplored candidate intersection in the sources reviewed.

Proposal: Reproduce a class-balanced replay Transformer NIDS, add a multi-signal secure replay admission controller (label consistency + representation support + anomaly risk + cohort persistence), quarantine ambiguous samples rather than discarding them, and periodically re-audit stored replay items. Evaluate against label flipping and a timing-feature backdoor.

Primary contribution target: improve the security-utility-plasticity trade-off - prevent poisoned evidence from becoming persistent replay memory without blocking legitimate emerging attacks from being learned.

Core metrics: macro-F1/per-class recall/FPR; forgetting/intransigence; poison admission and ASR; legitimate-novel rejection/learning; analyst-query budget; gate/update/inference overhead.

Dataset plan: CICIDS2017 for direct replication + LYCOS-IDS2017 or one UQ NetFlow V2 dataset for robustness/generalization.

6-credit output: baseline + continual pipeline + two controlled attacks + proposed defense + baselines/ablations + two dataset settings + reproducibility package; dashboard/edge device only as optional presentation work.

Review sentence

“We are not proposing another Transformer IDS. We are investigating how to make replay-based continual Transformer NIDS safe to update: specifically, how to stop poisoned or backdoored samples from becoming persistent replay memory without rejecting genuine new attacks that the model must learn.”

End of report.